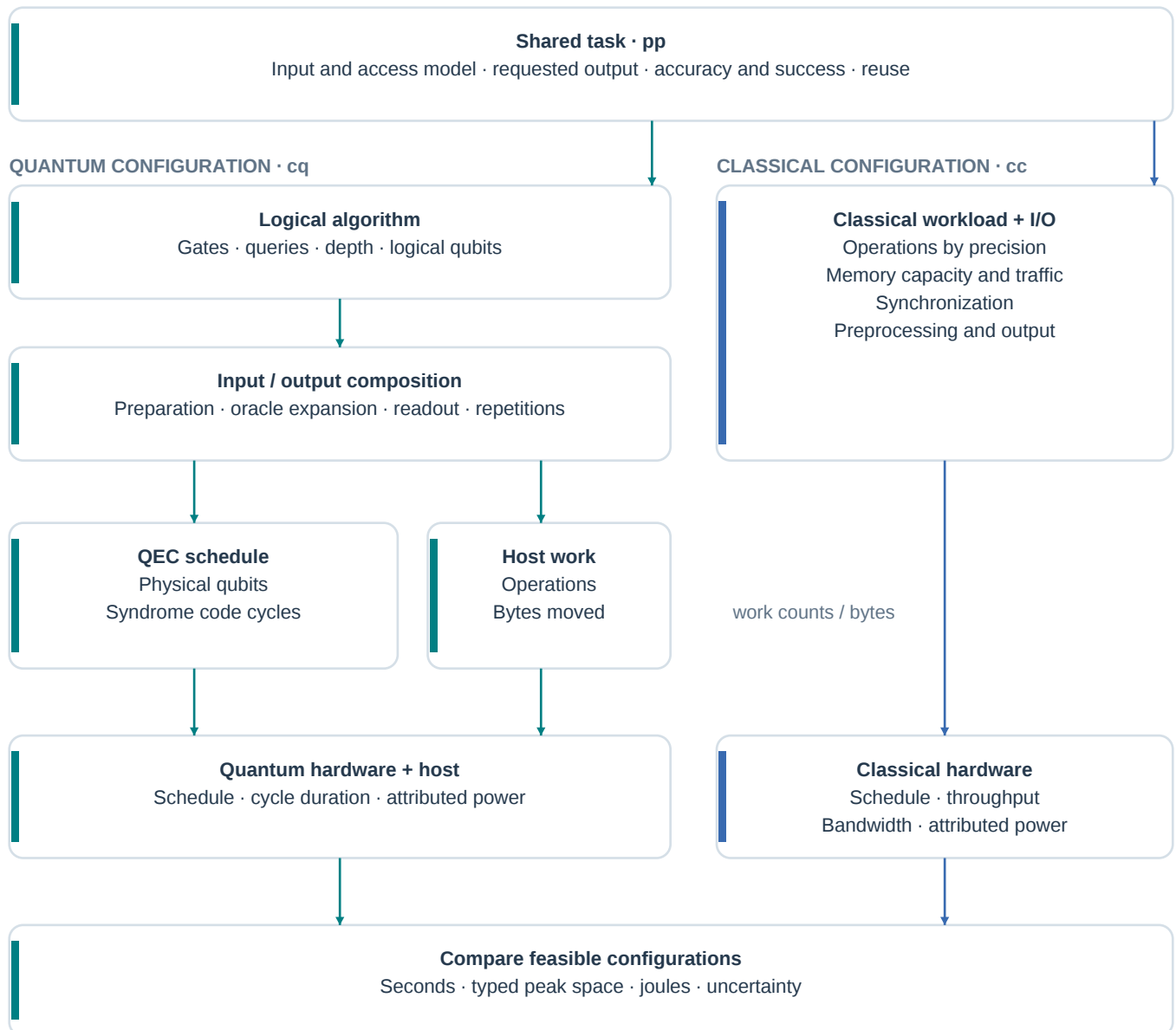


Compare the same answer, across the full computation.

A reusable framework for time, space and operational energy — with linear-system solvers as case studies.

The framework connects algorithmic work to physical execution through explicit interfaces. HHL, a modern quantum linear-system algorithm, and an eligible classical solver can each be evaluated against one shared task. The result is a conditional resource comparison, with feasibility and uncertainty attached.



QEC = quantum error correction. Host processing bypasses QEC and joins physical scheduling. Every branch includes the costs needed to make the requested output available.

Three inputs. Explicit contracts.

Keep the problem, algorithmic work and hardware performance separately identifiable.

Input	What it specifies
pp Problem parameters	Original matrix dimension N , matrix class, row/column sparsity and condition number κ ; input representation and location; right-hand side; output, accuracy, success and reuse. No scaling relation between N , κ and sparsity is implicit.
cq Quantum configuration	Algorithm, compilation and synthesis, input/output protocol, precision, error correction, factories, device and host hardware, limits and optimization policy.
cc Classical configuration	Solver and preconditioner, precision, input/output protocol, node and accelerator configuration, memory, network, limits and optimization policy.

$$F_q(pp, cq), F_c(pp, cc) \rightarrow (\text{time, space, energy, feasibility, uncertainty})$$

What each model hands to the next

Layer	Resource contract and units
Logical quantum	Gate counts in an exclusive basis; diagnostic oracle queries; depth in layers; peak logical qubits per invocation.
Input / output	Complete executable workload: preparation, implemented oracles, measurements, resets, repetition counts and host work.
Fault tolerance	Scheduled syndrome-extraction cycles and peak physical qubits, including factories, storage, routing and stalls; achieved workload failure bound.
Classical work	Operation counts by kind and precision; peak memory in bytes; traffic in bytes per named interface; synchronization.
Physical execution	Makespan in seconds; energy in joules; peak capacity by resource type; feasibility against hardware limits.

Composition without double counting

Each cost has one owner and a declared scope: setup, per output or batch. Apply repetition once. After an oracle is expanded into gates, its query count is diagnostic. After QEC produces a total cycle count, its repetition and code-distance factors are already included. Derivation links describe transformations, not extra costs to add.

Make the assumptions visible.

A resource estimate is meaningful only together with its output contract, schedule and system boundary.

01 Match the requested output and its reliability

A classical solution vector, a normalized quantum state and a scalar observable are different deliverables. Specify one output $y = f(x)$, its error metric and the probability of meeting it. Convert local state, residual or approximation errors to that output metric before combining them.

$$\Pr[d_{\text{out}}(\hat{y}, f(x)) \leq \varepsilon_{\text{out}}] \geq 1 - \delta_{\text{total}}$$

02 Charge setup and reuse consistently

Include required preprocessing, loading, oracle construction, factorization, preparation and extraction. Reuse setup only while its inputs remain valid. For B sequential outputs sharing one setup, amortized time or energy equals the batch total divided by B . Report first-output latency separately; peak memory and qubit capacity are never divided by B .

03 Separate work counts from elapsed time

Circuit depth, gate count and physical cycle count carry different meanings. Classical instruction counts also require a throughput, memory and synchronization model. Schedules determine overlap and peak occupancy. A T count alone does not fix runtime; processor frequency does not specify sustained floating-point performance. [2, 3]

04 Use a common operational energy boundary

$$E = \int P_{\text{attributed}}(t) dt \quad [\text{joules}]$$

Include the attributed electrical cost of quantum cooling, control, decoding and hosts, or classical compute, memory and networking, plus consistently allocated facility overhead. Power in W/qubit is a model parameter. Do not add cooling or facility factors to measurements that already include them. Declare gross or incremental energy; embodied energy is a separate metric.

05 Propagate uncertainty through the whole pipeline

Vary correlated inputs together, including conditioning, success probability, hardware errors, bandwidth and power. Recompute scheduling, QEC choices and feasibility under a declared fixed or reoptimized configuration policy. Keep unsupported assumptions distinct from measurements and rigorous bounds.

Interpretation: compare seconds and joules for the same workload and statistic. Report space as a vector of qubits and memory bytes. An advantage region requires both machines to be feasible and the uncertainty to support the comparison.

From one observable to resource cost.

An accounting illustration only: no hardware constants or numerical crossover are assumed.

Let $\mu = \langle x|O|x \rangle$ for normalized $|x\rangle$, with Hermitian O and outcomes in $[-1, 1]$. If each prepared state has trace-distance error at most η , the observable bias is at most 2η . Choose $2\eta + \epsilon_{\text{stat}} \leq \epsilon_{\text{out}}$. Independent accepted samples obey the Hoeffding bound [1, 4]:

$$M \geq \text{ceil}[2 \epsilon_{\text{stat}}^{-2} \ln(2/\delta_{\text{stat}})]$$

A trial resets and prepares the input, runs the solver and checks a success flag; let its gate vector be g_{try} . Each success adds observable readout and cleanup g_{obs} . For independent trials with constant success probability $p > 0$, an uncapped run needs M/p trials in expectation:

$$E[G] = (M/p) g_{\text{try}} + M g_{\text{obs}}$$

A deadline guarantee instead needs an attempt cap K with $\Pr[\text{Binomial}(K, p) < M] \leq \delta_{\text{retry}}$. Reserve K trials and M readouts. If c_{try} and c_{obs} include all QEC cycles and stalls, and τ_{cyc} is seconds per cycle, a serialized upper bound is:

$$\begin{aligned} t_q^{\text{ub}} &= t_{q,\text{setup}} + \tau_{\text{cyc}}(K c_{\text{try}} + M c_{\text{obs}}) + t_{q,\text{host}} \\ \delta_{\text{stat}} + \delta_{\text{retry}} + \delta_{\text{qec}} + \delta_{\text{other}} &\leq \delta_{\text{total}} \end{aligned}$$

For a classical solver of the same μ , let k iterations each require F_{it} operations and V_{it} bytes at a named interface. With sustained rates f_{eff} and b_{eff} , a deliberately serialized compute/transfer model gives:

$$t_c = t_{c,\text{setup}} + k(F_{\text{it}}/f_{\text{eff}} + V_{\text{it}}/b_{\text{eff}}) + t_{c,\text{output}}$$

For constant attributed run power, $E = E_{\text{setup}} + P_{\text{run}} t_{\text{run}}$. Keep setup outside the run window. Expected costs, reserved upper bounds and measured runtimes must remain clearly labeled when compared.

Primary references

- [1] Harrow, Hassidim & Lloyd (2009). Quantum algorithm for solving linear systems of equations. [arXiv:0811.3171v3](#).
- [2] Litinski (2019). A Game of Surface Codes. [Quantum 3, 128](#).
- [3] Williams, Waterman & Patterson (2009). Roofline. [CACM 52\(4\), 65–76](#).
- [4] Hoeffding (1963). Probability Inequalities for Sums of Bounded Random Variables. [JASA 58\(301\), 13–30](#).